Position-Conditioned Deformation Prediction with Unknown Material and Environmental Interactions

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Abstract

Predicting deformation from visual observations requires distinguishing an object's material response from the effects of its environment. Similar observed motions can result from different combinations of elasticity, fixture compliance, gravity, and spatial loading. We study position-conditioned prediction: a model observes a short sequence of object positions and prescribed tool poses, then predicts deformation under a new positional interaction without receiving the true force history. Our formulation separates object-attached material properties from environmental interaction fields and fits their latent parameters through implicitly differentiated nonlinear equilibrium. We develop a controlled benchmark that separates familiar positional motions from held-out interaction families and evaluates prediction before and after additional observations. Numerical reference checks and matched observation interfaces support comparisons with physical digital-twin methods and learned positional predictors. Across 42 held objects, the spatial inverse reduces new-interaction error by 43.1%, 78.3% and 68.9% in the familiar-parameter, material-shift and environmental-shift cohorts, respectively, relative to a positional neural baseline trained on all local interaction families. On nine spatial-transfer objects, one additional response reduces held-target error by 73.2%; sensitivity-based acquisition matches a strong fixed probe. The results demonstrate the value of inferring object-specific material and environmental explanations from a shared positional observation interface.

1. Introduction

Physical digital twins connect visual reconstruction to predictions about how an object responds to interaction. A useful twin should answer a positional query: if a gripper moves here or changes its orientation, how will the object deform? The reaction forces are consequences of the interaction, rather than necessarily available inputs. This distinction matters when deploying a model without calibrated force sensing.

Material properties alone do not determine the observed response. Fixtures may deform, constraints may be only partially visible, and distributed forces may act throughout the volume. A model that explains an observed trajectory by changing material stiffness can reproduce that trajectory while predicting a different interaction incorrectly. The resulting ambiguity is especially consequential outside the interaction patterns used for reconstruction.

We investigate a spatially factored model of this problem. Material coordinates index the object, while environmental interactions are evaluated at the current spatial positions and tool poses. This representation makes the action interface positional and allows forces to emerge from the inferred interaction model. We evaluate generalization across explicitly declared interaction groups rather than treating additional noisy observations of the same motion as independent out-of-distribution evidence.

We formulate position-conditioned inverse mechanics with distinct material and spatial interaction fields, and derive a whole-object translation test that distinguishes their physical explanations. We develop an independently refined simulation benchmark with material, environmental-parameter, and force-law shifts. Finally, we evaluate matched-information prediction and one-observation adaptation against learned and physical component baselines.

Our study builds on complementary advances in physical Gaussian representations and visual system identification. PhysGaussian [7] couples continuum simulation and Gaussian rendering, while PhysTwin [3] reconstructs visual and physical object models from interaction videos. These systems establish strong tools for building and using digital twins. Our experiments target the additional question of how unknown environmental interactions affect positional generalization, and how limited new observations can resolve those ambiguities.

2. Related Work

Differentiable mechanics. DiffPD [2] develops efficient differentiable soft-body simulation through implicit projective dynamics, including contact models and applications to system identification and control. Our inverse model uses

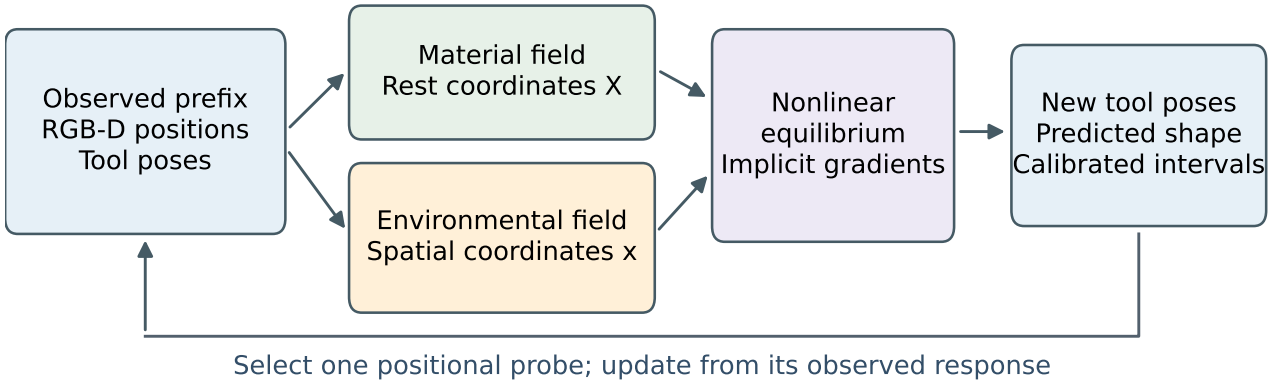


Figure 1. Position-conditioned inverse mechanics. Observed object positions constrain material and environmental explanations. New tool poses are converted to interaction through the inferred model; true force histories are not supplied. Local predictive sensitivities support selecting an additional observation.

equilibrium sensitivities to fit settled observations. We focus on separating object-attached material variation from spatial environmental loading, and on testing the resulting explanations under new positional queries.

Physical visual representations. PhysGaussian [7] integrates 3D Gaussian rendering with material-point simulation. Its representation supports a broad range of material behaviors and prescribed scene interactions. PhysTwin [3] combines spring-mass simulation, geometric reconstruction, and Gaussian appearance modeling to infer deformable twins from sparse interaction videos. We compare compatible physical prediction components under a common observation interface and identify adaptations explicitly; our controlled experiments do not substitute for an evaluation of every capability of these complete systems.

Learning physical response. Spring-Gaus [10] identifies elastic spring-mass models from video and evaluates future simulation under changed initial states and environmental parameters. PhysDreamer [9] uses video-generation priors to estimate spatial material fields and synthesize plausible interactions. UniPhy [5] learns a shared latent-conditioned constitutive model across elastic, plastic, granular, and fluid materials, then optimizes a latent from observed positions for re-simulation under new conditions. PhysHO [4] derives human actuation from pose trajectories through a PD controller coupled to MPM and learns spatial residual constitutive models. It demonstrates position-driven physical prediction under new human actions. These results establish that material inference and interaction generalization are already important capabilities of existing work. Our controlled study addresses their interaction with unknown environmental explanations and the selection of additional positional observations. ParticleGS [6] learns latent particle dynamics from multi-view videos with a neural ODE and decodes the evolved states into Gaussian deformations for future rendering. Its temporal extrapolation task complements our counterfactual positional queries: we prescribe new tool poses and evaluate the resulting settled geometry.

Acquiring informative interactions. AuRA [8] automates robot interaction and data preparation for deformable digital-twin reconstruction. Its focus on efficient acquisition complements our study of uncertainty in the physical explanation of an observed interaction. We distinguish initial generalization from adaptation after collecting additional observations, and report acquisition and computational costs separately.

3. Problem Formulation

Let $X \in \Omega_0$ denote a material point in a canonical object, and let $x(X, a) = X + u(X, a)$ be its spatial position under a positional action a. The action specifies tool translations and rotations. An observed prefix $\mathcal{D} = \{a_j, y_j\}_{j=1}^{m}$ contains partial position observations obtained through the visual pipeline. The prediction task is to estimate $u(X, a_*)$ for an unseen action a_*, together with predictive uncertainty when supported by the fitted model.

The latent variables are divided into material parameters θ_m and environmental parameters θ_e. True reaction forces and simulator material labels are excluded from the primary

model inputs. Privileged experiments that supply such labels are reported separately. Known geometry and camera calibration are explicit assumptions of the controlled protocol.

4. Spatially Factored Inverse Mechanics

4.1. Position-driven equilibrium

We model settled deformation as a stationary point of total potential energy,

$$u^*(a,\theta) = \arg\min_u \Pi(u,a,\theta), \tag{1}$$

$$\begin{aligned}\Pi = &\int_{\Omega_0} W(F;\theta_m)\,dX \\ &+ \sum_k \frac{1}{2}\int_{\Gamma_k} \kappa_k(X)\|u - d_k(a)\|^2\,dS \\ &- \int_{\Omega_0} \rho g^\top u\,dX + \int_{\Omega_0} \phi(X+u;\theta_e)\,dX. \end{aligned} \tag{2}$$

Here $F = I + \nabla_X u$, $d_k(a)$ is the displacement prescribed by a rigid tool pose, and κ_k represents compliant coupling to that tool or fixture. Gravity is a body force; surface constraints and spatial loading retain their distinct physical roles. The potential ϕ is evaluated in spatial coordinates, so its force changes as material moves through the environment.

We use a compressible Neo-Hookean energy,

$$\begin{aligned}W(F) = &\frac{\mu}{2}(\operatorname{tr}(F^\top F) - 3) - \mu \log J \\ &+ \frac{\lambda}{2}(\log J)^2, \qquad J = \det F. \end{aligned} \tag{3}$$

The simulator enforces a positive-Jacobian guard and checks equilibrium residuals. Reference solutions use quadratic finite elements with spatial and quadrature refinement. Displacement qualification does not imply convergence of every derived stress or traction quantity.

4.2. Material and environmental fields

The material field is attached to the reference object. In the controlled study,

$$E(X) = E_0 \exp[h\,(b_1(X) - b_2(X))]\,, \tag{4}$$

where b_i are fixed radial basis functions along the block's length. The coefficient h controls spatial stiffness variation. The environmental potential is a spatial Gaussian, $\phi(x) = A|\Omega_0|^{-1}\exp(-\|x-c\|^2/(2s^2))$, with inferred energy amplitude A, center c, and width s. Its force is $-\nabla_x\phi(x)$ and changes as the object traverses the field. Two compliant surface footprints couple the object to the position-controlled tools. Each footprint is a Gaussian in the end-face coordinates, tapered to zero at the perimeter. We infer its total stiffness K_k and select among three footprint-layout hypotheses. With nonnegative footprint $w_k(X)$,

$$\kappa_k(X) = \frac{K_k w_k(X)}{\int_{\Gamma_k} w_k(X)\,dS}. \tag{5}$$

The normalization separates contact location from total coupling stiffness. The smooth finite footprint also avoids an abrupt traction transition at an idealized half-face attachment.

Gravity has known magnitude and unknown direction in the observation frame. Density, Poisson ratio, canonical geometry, and tool kinematics are shared known quantities. Thus the study isolates uncertainty in stiffness, attachment, and environmental loading. Material OOD tests change elastic coefficients and heterogeneity within this constitutive family.

Why spatial transfer distinguishes the explanations. Consider translating both tools by the same vector t. With material-indexed body loads, gravity, and the compliant attachments above, $u(a+t) = u(a) + t$ is an equilibrium whenever $u(a)$ is: deformation gradients and relative tool displacements are unchanged. A spatially varying load generally breaks this symmetry. Let T repeat a translation at every node and let K_ϕ be the spatial-potential contribution to the tangent. Implicit differentiation gives

$$\frac{\partial u^*}{\partial t} = T - K^{-1}K_\phi T. \tag{6}$$

Thus a whole-object positional query directly exposes the spatial curvature of the environmental potential. This local identity assumes a nonsingular equilibrium and applies to the stated conservative model. It motivates the transfer experiment and the matched material-indexed-load ablation.

4.3. Inference from positional observations

For an observation operator H_j, we fit latent parameters by minimizing a position discrepancy,

$$\mathcal{L}(\theta) = \sum_j \|H_j\hat{u}(a_j,\theta) - y_j\|^2_{\Sigma_j^{-1}} + \|z - z_0\|^2_\Lambda. \tag{7}$$

Here z denotes scaled latent coordinates, with log coordinates for positive stiffnesses and field width. Weak regularization constrains poorly identified directions. We screen the three attachment layouts using a coarse model and refine layouts within one normalized loss unit of the best candidate. Parameter bounds and initialization are selected on development cases before the held-out evaluation. The predictor receives the positional action, not the forces computed internally by the equilibrium model.

Let $R(u, \theta, a) = \partial\Pi/\partial u$ and $K = \partial R/\partial u$. At a nonsingular equilibrium, an adjoint solve gives

$$K^\top v = \frac{\partial \mathcal{L}}{\partial u}, \qquad \frac{d\mathcal{L}}{d\theta} = \frac{\partial \mathcal{L}}{\partial \theta}\bigg|_u - v^\top \frac{\partial R}{\partial \theta}. \tag{8}$$

We verify the implementation against an independent sparse assembly and finite differences of the equilibrium solution.

Reducing forward discretization bias. A coarse forward model can explain its own numerical error by changing material or environmental parameters. We therefore combine equilibrium predictions at two resolutions,

$$\hat{u} = \frac{9u_3 - 4P_{2\to3}u_2}{5}, \tag{9}$$

where u_n uses $4n \times n \times n$ quadratic elements and $P_{2\to3}$ interpolates the coarse field. This removes a leading quadratic resolution term when that error model applies. We assess the correction on development oracle cases, differentiate both equilibrium solves, and retain a positive-Jacobian check on the corrected field. The corrected field approximates the continuum response; remaining discrepancy is included in predictive calibration.

4.4. Predictive sensitivity and probing

Let $J_a = \partial\hat{u}(a, \hat{\theta})/\partial z$ denote the positional sensitivity at the fitted explanation. We construct a regularized local covariance

$$C = \left(J_{\mathcal{D}}^\top \Sigma^{-1} J_{\mathcal{D}} + \Lambda\right)^{-1}, \tag{10}$$

where Λ encodes weak parameter-scale regularization. This local approximation describes sensitivity around a fitted explanation, conditioned on its selected attachment layout. A separate object-level calibration stage sets the predictive interval scale.

For a candidate probe a, observing its visible points yields an approximate covariance $C_a = (C^{-1} + J_a^\top H_a^\top \Sigma^{-1} H_a J_a)^{-1}$. We choose the probe that maximizes the reduction in target predictive variance,

$$a^* = \arg\max_a \operatorname{tr}\left[J_* \left(C - C_a\right) J_*^\top\right]. \tag{11}$$

Probe selection uses predicted sensitivities, without observing candidate responses. The selected response is then acquired and the inverse problem is refitted. A separate target interaction remains unobserved throughout adaptation.

Object-level calibration. For each calibration object, we take the 90th percentile of absolute held-action coordinate errors divided by $\sqrt{\sigma^2 + \epsilon^2}$, where $\sigma^2 = \operatorname{diag}(J_a C J_a^\top)$ and $\epsilon = 20\,\mu$m is a fixed scale floor. Following split conformal calibration [1], the order statistic at rank $\lceil 0.9(n+1) \rceil$ of the n object scores determines a common interval multiplier. Under object exchangeability, this construction targets at least 90% coordinate coverage for 90% of objects, marginally over calibration and test draws. For spatial-transfer experiments, we fit separate multipliers before acquisition and after each acquisition policy on 12 independent calibration objects, using the same held-target protocol as evaluation. This separates covariance contraction from demonstrated post-acquisition coverage. The benchmark fixes balanced attachment-layout counts, and the shifted cohorts additionally change their parameter distributions. We therefore report coverage empirically for this design, rather than assert an exchangeability guarantee for every cohort.

4.5. Generalization and adaptation

We distinguish three questions: prediction under a new action before its response is observed; calibration of uncertainty about that response; and prediction after a limited adaptation budget. Each object's environment remains fixed during these experiments; positional actions change how the object samples that environment. Environmental OOD changes the distribution of fields across objects, whose prefixes remain observable. Completely unobserved time-varying changes in external loading are outside this protocol.

5. Experimental Protocol

Geometry and observations. The controlled object is a $240 \times 80 \times 80$ mm deformable block with known canonical geometry, density $625\,\text{kg/m}^3$, and Poisson ratio 0.35. Two position-controlled tools couple to unknown smooth Gaussian attachment footprints. The observed prefix contains three settled configurations: nominal tool poses, a 6 mm right-tool translation along y, and a 6 mm translation along z. A 640×480 rendered RGB-D marker pipeline provides the same 27 visible point positions to all methods. Known synthetic marker IDs establish correspondence; depth has independent 0.3 mm Gaussian noise and image sampling introduces pixel quantization. The positional objective uses a common 0.3 mm coordinate scale. No force, reaction, material, or attachment labels are supplied. Predictions are evaluated at 81 canonical query points, with visible and hidden-surface subsets reported separately.

Cohorts and shifts. The protocol declares nine development objects, twelve calibration objects, and three disjoint test cohorts: eighteen objects from the development parameter ranges, twelve with extrapolated elastic coefficients and spatial heterogeneity, and twelve with shifted environmental-field amplitude, center, width, and gravity direction. Material coefficients within development range from 8 to 40 kPa, and total attachment stiffnesses from

1500 to 15000 N/m. Each object has a new 3 mm familiar-direction query, five new positional interaction patterns, and a doubled-amplitude query. New patterns include stretching, opposed tool motion, twisting, bending, and diagonal translation. An additional transfer challenge translates both tools together through the fixed spatial field. These are declared interaction and parameter shifts; the reference material law remains Neo-Hookean.

Numerical references. Reference geometry uses quadratic finite elements and independent spatial and quadrature refinement. Every action must have a displacement RMS change below $20\,\mu$m, equilibrium residual below 10^{-9} N, positive deformation Jacobians above 0.1, and reaction balance below 10^{-7} N. Maximum-coordinate refinement changes are retained separately, since localized attachment-edge behavior converges more slowly. Acceptance in an RMS norm does not certify pointwise stress convergence. The inverse predictor uses a smaller discretization and an independently checked dense tangent implementation. A development-only sharp-attachment pilot motivated the finite smooth footprint: its localized compression reached the Jacobian guard under refinement, whereas the smooth version retained substantial nonrigid deformation and passed the original numerical thresholds.

Deformation content. All 87 reference fields pass the declared criteria: 63 independent development, calibration, and test objects, plus 24 spatial-transfer fields that reuse their physical parameters. The largest spatial-refinement RMS change is 19.34 μm, the largest quadrature-refinement change is 0.278 μm, and the smallest deformation Jacobian is 0.727. These checks qualify the displacement references used for evaluation.

Across all 42 test objects, the median object-average nonrigid coordinate RMS over the five new interaction patterns is 4.64 mm, with an object range of 2.24–12.38 mm. Relative to the already-deformed nominal configuration, the median nonrigid change induced by those actions is 0.89 mm (range 0.64–1.19 mm), after removing the best rigid transform between configurations. The benchmark therefore contains substantial shape change at both the configuration and interaction levels.

Comparisons. The physical baselines use official PhysTwin spring-mass and PhysGaussian material-point simulation components with documented adapters for the shared positional interface. They retain their own constitutive representations. PhysTwin learns per-spring parameters; the MPM adapter fits material variation, compliant tool coupling, and gravity direction. We tune initialization, discretization, and settling on development cases. A positional neural predictor is trained on a separate synthetic object bank, with an ID-action training variant and a separately reported variant exposed to all local action families. Controlled ablations fix environmental loading to its nominal value or freeze the spatial force at each material point's canonical position. The main predictor and neural comparisons cover all 42 declared test objects. The more expensive physical comparisons use the first six objects in each test cohort, giving two objects per attachment layout and an 18-object matched subset selected before test inference. Full Gaussian appearance reconstruction is outside this component benchmark.

Metrics and adaptation. The primary error is coordinate RMS displacement error,

$$e = \sqrt{\frac{1}{3N}\sum_{i=1}^{N} \|\hat{u}_i - u_i\|_2^2}. \tag{12}$$

We also report Euclidean point RMSE, hidden-surface error, and residual motion after removing the best rigid transform. Incremental error compares $\hat{u}(a) - \hat{u}(a_0)$ against $u(a) - u(a_0)$, separating the predicted response to the new action from static shape-completion error. Aggregation treats the object as the independent unit. Paired comparisons resample objects rather than frames or points. Probing compares a fixed $+40$ mm y translation, a reproducible random action, and the sensitivity-based policy, all with one additional acquired response and a separate -40 mm y target. Twelve independent calibration objects set separate pre- and post-acquisition interval scales; the first nine familiar-parameter test objects evaluate transfer. A material-indexed-load ablation receives exactly the probe response selected for the spatial model. Before acquisition, the two neural baselines also predict all nine transfer objects; physical baselines reuse their original fits on the first six objects, with no additional observed responses.

6. Prediction Results

Generalization across interactions and physical shifts. Table 1 reports all 42 test objects. With only three positional observations, the spatial inverse achieves new-motion coordinate RMSE of 0.068, 0.087 and 0.067 mm for familiar parameters, material shifts and environmental shifts, respectively. Relative to the neural predictor trained on all local interaction families, the reductions are 43.1%, 78.3% and 68.9%. The paired mean advantages are 0.051, 0.315 and 0.149 mm, with 95% object-bootstrap intervals $[0.037, 0.066]$, $[0.212, 0.448]$ and $[0.066, 0.276]$ mm. Since all-action neural training includes these interaction families on disjoint objects, this comparison tests the benefit of per-object inference beyond action-family exposure.

Method	Familiar parameters		Material shift		Environment shift	
	New motions	2× amplitude	New motions	2× amplitude	New motions	2× amplitude
Spatial inverse (ours)	0.068	0.071	0.087	0.089	0.067	0.073
Neural (ID actions)	1.528	0.130	1.551	0.423	1.488	0.218
Neural (all actions)	0.119	0.119	0.402	0.377	0.216	0.223

Table 1. Coordinate RMSE (mm), lower is better. All 42 declared test objects. Object-level bootstrap intervals and familiar-direction queries are provided in the accompanying report.

The spatial inverse retains low error at doubled action amplitude (0.071–0.089 mm). On material-shift objects, its incremental new-motion error is 0.020 mm versus 0.195 mm for all-action training, indicating improved response prediction beyond reconstruction of the nominal shape. These material shifts remain within the reference constitutive family.

Matched physical component comparison. Table 2 reports all 18 predeclared objects, six per cohort. New-motion errors for the spatial inverse, MPM and PhysTwin adaptations are $0.075/0.310/0.860$ mm on familiar parameters, $0.091/0.364/1.495$ mm on material shifts, and $0.065/0.303/1.061$ mm on environmental shifts. Paired advantages over MPM are 0.235, 0.273 and 0.238 mm, with 95% object-bootstrap intervals $[0.169, 0.302]$, $[0.148, 0.403]$ and $[0.166, 0.305]$ mm. The material-shift advantage over PhysTwin is 1.404 mm, with interval $[0.903, 2.017]$ mm.

Incremental-response errors on material shifts are $0.021/0.117/0.639$ mm, and on environmental shifts $0.031/0.208/0.655$ mm, in the same method order. Thus the differences extend beyond nominal-shape reconstruction. The supplement reports every cohort's incremental errors. These comparisons use the common three-observation positional interface, development-selected settings, and each adapted component's own constitutive representation.

Accounting for the environment. On the same six environmental-shift objects, the nominal-environment ablation fixes gravity downward and omits the spatial field. Its new-motion error is 3.050 mm, versus 0.065 mm for the spatial inverse. Incremental-response error is 0.281 versus 0.031 mm, so the benefit extends beyond correcting the nominal shape. This compares the declared ablation settings, including their regularization choices. With the same constitutive model, inverse resolution and regularization, material-indexed loading gives 0.075 mm on these environmental-shift objects; the spatial model has paired advantage 0.010 mm with 95% interval $[0.001, 0.019]$ mm. On six familiar-parameter objects, the two are nearly tied (0.075 mm spatial, 0.074 mm material-indexed). On the six material-shift objects, errors are 0.091 and 0.096 mm, respectively, with paired advantage 0.005 mm and interval $[-0.0001, 0.0114]$ mm. Thus all three local-query cohorts show only a modest distinction between these coordinate-frame explanations.

Predictive interval coverage. With one interval scale fixed on twelve independent calibration objects, average held-coordinate coverage is 98.3%, 95.9% and 98.7% for familiar parameters, material shifts and environmental shifts, respectively. Mean half-widths are 0.169, 0.168 and 0.181 mm. At least 90% of held coordinates are covered for $18/18$, $10/12$ and $12/12$ objects, respectively; the lowest material-shift coverage is 88.9%. These results use all seven held local actions per object and the unchanged calibration scale. They measure predictive uncertainty before acquisition, separately from material-parameter identifiability and post-probe performance. The supplement reports coverage and width together.

Whole-object translation before acquisition. Across six translations on all nine declared transfer objects, coordinate RMSE is 0.183 mm for the spatial inverse and 0.141 mm for material-indexed loading, versus 11.436 and 16.514 mm for all-action and ID-action neural training. Neither neural variant was trained on these large global translations. The paired spatial-model advantage over material-indexed loading is -0.042 mm with 95% interval $[-0.110, 0.039]$ mm. On the separately held -40 mm y target, errors are 0.314 and 0.248 mm, respectively. Thus mechanics supports strong positional transfer, but three local observations do not establish a predictive advantage for the spatial coordinate choice before acquisition. The independently calibrated pre-acquisition intervals cover 81.9% of held-target coordinates with mean half-width 0.373 mm; $5/9$ objects reach 90% coverage. The post-probe comparison evaluates an additional response through prediction error, coverage and width together.

On the six predeclared physical-baseline transfer objects (Table 4), translation errors are 0.194, 0.368 and 0.804 mm for the spatial inverse, MPM and PhysTwin adaptations, respectively. Paired advantages over MPM and PhysTwin are 0.175 and 0.610 mm, with 95% intervals $[0.076, 0.274]$

Method	Familiar parameters		Material shift		Environment shift	
	New motions	2× amplitude	New motions	2× amplitude	New motions	2× amplitude
Spatial inverse (ours)	0.075	0.084	0.091	0.091	0.065	0.078
Neural (ID actions)	1.553	0.141	1.509	0.487	1.525	0.270
Neural (all actions)	0.127	0.130	0.471	0.449	0.297	0.300
PhysTwin core adaptation	0.860	0.816	1.495	1.381	1.061	0.911
PhysGaussian MPM adaptation	0.310	0.341	0.364	0.352	0.303	0.327
Material-indexed load	0.074	0.076	0.096	0.116	0.075	0.098
Nominal environment	1.024	0.945	2.577	2.375	3.050	2.901

Table 2. Coordinate RMSE (mm), lower is better. Matched physical comparison on 18 objects selected before evaluation (six per cohort). Native physical components use our shared-interface adapters. Object-level bootstrap intervals and familiar-direction queries are provided in the accompanying report.

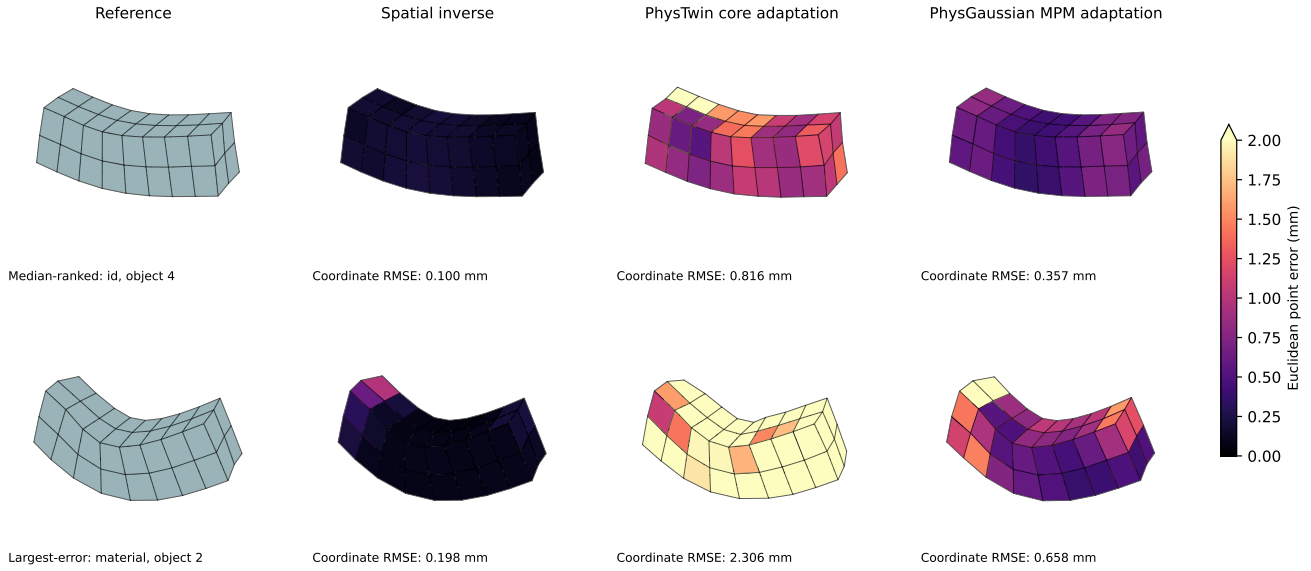


Figure 2. Actual-scale geometry and prediction error for the upper-median and largest-error objects among the 18 matched cases, ranked by the spatial inverse's mean error over seven held actions. Each row shows that object's highest-error held action for the spatial inverse. All methods are shown on the same case and action. Color denotes Euclidean point error; reported panel values use coordinate RMSE.

and $[0.423, 0.799]$ mm. Held-target errors are 0.330, 0.533 and 0.830 mm, respectively. Every model reuses its original three-observation fit. This subset comparison uses no newly acquired responses and is separate from the nine-object acquisition study.

Using the same additional positional response. The complete nine-object matched-frame comparison separates the value of the environmental representation from probe selection. After observing the same additional response, held-target error falls from 0.314 to 0.084 mm for the spatial model, while material-indexed loading changes from 0.248 to 0.297 mm. The post-acquisition paired advantage is 0.213 mm with 95% object-bootstrap interval $[0.057, 0.409]$ mm. The spatial model can use the new observation to improve predictions across the fixed world field; a material-indexed explanation cannot reproduce that position dependence. All nine sensitivity-based choices equal the fixed $+40$ mm y probe. This result supports the representation's use of additional evidence, while the adaptive and fixed policies make identical acquisitions on these objects.

One-response acquisition and predictive uncertainty. Table 5 reports all nine objects; the supplement visualizes error, coverage, interval width and the matched-frame comparison. Fixed and sensitivity-based probing reduce held-target error by 73.2%, from 0.314 to 0.084 mm; reproducible random probing gives 0.163 mm. Their paired advantage over random probing is 0.079 mm with 95% object-bootstrap interval $[0.013, 0.151]$ mm. The fixed probe is a strong mirror of the held target, and the sensitivity pol-

Method	Anisotropic harmonic field		Two-Gaussian field	
	New motions	Spatial transfer	New motions	Spatial transfer
Spatial inverse (ours)	0.082	0.392	0.068	0.267
Material-indexed load	0.095	0.665	0.068	0.249
Neural (ID actions)	1.568	16.097	1.448	16.459
Neural (all actions)	0.136	11.041	0.139	11.413
PhysTwin core adaptation	0.733	0.945	0.864	0.773
PhysGaussian MPM adaptation	0.307	0.742	0.269	0.350

Table 3. Unseen environmental force laws: coordinate RMSE (mm) on a separately predeclared six-object extension, three objects per law. All methods use their frozen settings and the same original three observations. Material coefficients remain unknown but are drawn within the familiar range. No model is retrained on these force laws.

Method	Translations	Target
Spatial inverse (ours)	0.194	0.330
Neural (ID actions)	16.528	14.298
Neural (all actions)	11.390	7.534
PhysTwin core adaptation	0.804	0.830
PhysGaussian MPM adaptation	0.368	0.533
Material-indexed load	0.169	0.316

Table 4. Spatial-transfer coordinate RMSE (mm) before acquisition, on six matched objects. Every model reuses its original three-observation fit. The held target is the -40 mm y translation.

Acquisition	Error	Coverage	Half-width
Before	0.314	81.9%	0.373
Fixed	0.084	92.0%	0.157
Random	0.163	88.1%	0.238
Adaptive	0.084	92.0%	0.157

Table 5. Held spatial target before and after one acquired response on nine objects. Error is coordinate RMSE and widths are in mm. Each policy uses a separately fitted scale from twelve calibration objects; coverage and width are assessed together.

icy selects it for every object, yielding identical predictions. The experiment therefore demonstrates useful one-response adaptation and an advantage over these random draws, without an advantage over the fixed choice.

With policy-specific scales frozen on twelve independent calibration objects, fixed/adaptive probing raises mean coordinate coverage from 81.9% to 92.0% while reducing mean interval half-width from 0.373 to 0.157 mm. Random probing gives 88.1% coverage and 0.238 mm half-width. Six of nine objects reach 90% coordinate coverage after each policy, compared with five before acquisition. Thus narrower intervals accompany improved aggregate coverage and prediction, although object-level coverage remains uneven in this small transfer cohort.

Prediction under unseen force laws. Table 3 reports the complete six-object extension: three anisotropic harmonic fields and three signed two-Gaussian fields, evaluated with frozen settings and the original three observations. Spatial-inverse new-motion errors are 0.082 and 0.068 mm, compared with $0.136/0.139$ mm for all-action neural training, $0.307/0.269$ mm for MPM and $0.733/0.864$ mm for PhysTwin. On whole-object translations, spatial-inverse errors are $0.392/0.267$ mm versus $0.665/0.249$ mm for material-indexed loading: the spatial representation helps under harmonic loading, while the two-Gaussian cases slightly favor material-indexed loading. This retains the distinction between physical transfer and a universal coordinate-frame advantage.

Across all six objects, paired advantages over MPM are 0.213 mm on new motions and 0.217 mm on translations, with descriptive 95% object-bootstrap intervals $[0.186, 0.235]$ and $[0.086, 0.372]$ mm. These small, predeclared groups support transfer under the tested force-law misspecification; the report retains all per-object results.

7. Discussion

Position-conditioned inverse mechanics predicts new interactions from a small observed prefix while treating material, coupling and environmental loading as unknown. The matched-frame acquisition experiment shows how an additional positional response can distinguish explanations that are difficult to separate through local observations alone.

The reference and proposed predictor share a Neo-Hookean constitutive family, while the native physical baselines retain their original constitutive representations. The comparison therefore measures these adapted predictors within the declared family. The matched material-indexed-load ablation isolates the coordinate-frame choice using the same constitutive model and inverse solver. Known geometry, synthetic correspondence, and settled conservative mechanics make this a controlled study of physical prediction; dynamic impact and release, real visual correspondence, and broader constitutive families are natural extensions.

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